

MATH563: Convex Optimization

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Abstract

1 Preliminaries

Definition 1 (Affine set). A set $C \subseteq \mathbb{E}$ is called an affine set if for all $x, y \in C$ and $\lambda, \mu \in \mathbb{R}$, we have

$$\lambda x + \mu y \in C. \quad (1)$$

Definition 2 (Affine hull). The affine hull of a set $Q \subseteq \mathbb{E}$ denoted by $\text{aff}(Q)$, is the intersection of all affine sets that contain Q .

Affine hull is the smallest affine set that contains Q .

Definition 3 (Relative interior and boundary). The relative interior of a set $Q \subseteq \mathbb{E}$, denoted by $\text{ri}(Q)$, is the interior of Q relative to its affine hull, i.e.

$$\text{ri}(Q) = \{x \in Q : \exists \delta > 0, B(x, \delta) \cap \text{aff}(Q) \subseteq Q\}. \quad (2)$$

The relative boundary of Q is defined as $\text{rb}(Q) = \text{cl}(Q) \setminus \text{ri}(Q)$.

Definition 4 (Epigraph). The epigraph of a function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is the set

$$\text{epi}(f) = \{(x, \alpha) \in \mathbb{E} \times \mathbb{R} : f(x) \leq \alpha\}. \quad (3)$$

Relative interior is weaker than interior since it's easier to be satisfied, as $B_\varepsilon \cap \text{aff} Q \subset B_\varepsilon$. We have $\text{int}(Q) \subseteq \text{ri}(Q)$ and $\text{rb}(Q) \subseteq \text{bd}(Q)$.

1.1 Lower semicontinuity

Definition 5 (Lower limit). For $f : \mathbb{E} \rightarrow \mathbb{R}$, and $\bar{x} \in \mathbb{E}$, we define

$$\liminf_{x \rightarrow \bar{x}} f(x) = \sup_{\delta > 0} \inf_{x \in B(\bar{x}, \delta)} f(x) \quad (4)$$

$$= \inf\{\alpha \in \mathbb{R} : \exists x^k \rightarrow \bar{x} \text{ s.t. } f(x^k) \rightarrow \alpha\}. \quad (5)$$

as the lower limit of f at $\bar{x}$.

Definition 6 (Continuity notions). Let $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and $\bar{x} \in \mathbb{E}$, then f is said to be lower semicontinuous at $\bar{x}$ if $f(\bar{x}) \leq \liminf_{x \rightarrow \bar{x}} f(x)$.

Alternatively, f is lsc at $\bar{x}$ iff there does not exist a sequence $\{x^k\} \rightarrow \bar{x}$ such that $f(x^k) < f(\bar{x})$.

Definition 7 (Closure of function). For $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$, the closure of f is the function $\text{clf} : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ defined by

$$\text{clf}(\bar{x}) = \liminf_{x \rightarrow \bar{x}} f(x). \quad (6)$$

As $\liminf_{x \rightarrow \bar{x}} f(x) \leq f(\bar{x})$, we always have $\text{clf} \leq f$. We also have $\text{epi}(\text{clf}) = \text{cl}(\text{epi} f)$. That said, epigraph and closure commute with each other.

Proposition 1 (Characterization of lsc). Let $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$. Then TFAE:

1. f is lsc on $\mathbb{E}$
2. $\text{epi}(f)$ is closed
3. $\text{lev}_{\leq \alpha} f$ is closed for all $\alpha \in \mathbb{R}$

Corollary 1 (lsc of indicator function). Let $C \subseteq \mathbb{E}$, its indicator function δ_C is proper, lsc iff C is nonempty and closed.

1.2 Optimization problems

We always have

$$\inf_C f = -\sup_C(-f), \quad \sup_C f = -\inf_C(-f). \quad (7)$$

Definition 8 (Minimizer). Let $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$ and $C \subseteq \mathbb{E}$. Then $\bar{x} \in C \cap \text{dom}(f)$ is called

1. local minimizer of f over C if

$$\exists \varepsilon > 0 : f(\bar{x}) \leq f(x), \forall x \in C \cap B(\bar{x}, \varepsilon). \quad (8)$$

2. global minimizer of f over C if

$$f(\bar{x}) \leq f(x), \forall x \in C. \quad (9)$$

When minimizing a proper function $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$, we have:

$$\arg \min_{\mathbb{E}} f \neq \emptyset \implies \inf_{\mathbb{E}} f \in \mathbb{R}. \quad (10)$$

but not the reverse.

Theorem 1 (Existence of minima). Let $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$ be proper, lsc, and level bounded. Then $\arg \min f \neq \emptyset$ and $\min f = \inf f \in \mathbb{R}$.

2 Convex Sets

Definition 9 (Convex sets). A set $C \subseteq \mathbb{E}$ is called convex if for all $x, y \in C$ and $\lambda \in [0, 1]$, we have

$$\lambda x + (1 - \lambda)y \in C. \quad (11)$$

Example of convex sets includes: subspace of $\mathbb{E}$, Minkowski sum of convex sets, hyperplanes, affine sets, intersection of convex sets, linear image and preimage of convex sets, half spaces, interval (closed, open, half-open).

2.1 Projection on convex sets

Definition 10 (Projection on a set). Let $S \subset \mathbb{E}$ be nonempty and $x \in \mathbb{E}$, then we define the projection of x on S by

$$P_S(x) = \arg \min_{y \in S} \|x - y\| \quad (12)$$

Now we give sufficient conditions for this subset to be nonempty and unique.

Lemma 1. Let $x \in \mathbb{E}$ and $S \subseteq \mathbb{E}$ be nonempty. Then the following hold:

1. If S is closed, then $P_S(x) \neq \emptyset$.
2. If S is convex, then $P_S(x)$ has at most one element.

Corollary 2 (Projection on closed, convex set). Let $C \subset \mathbb{E}$ be nonempty, closed, and convex. Then $P_C(x)$ is a mapping from $\mathbb{E}$ to C with $x = P_C(x)$ iff $x \in C$. That said, projection onto a closed convex set is unique.

The following theorem gives an important characterization of the projection on a closed convex set in terms of variational inequality.

Theorem 2 (Projection Theorem). Let $C \subseteq \mathbb{E}$ be nonempty, closed, and convex, and let $x \in \mathbb{E}$. Then $\bar{v} = P_C(x)$ iff

$$\langle x - \bar{v}, v - \bar{v} \rangle \leq 0, \quad \forall v \in C. \quad (13)$$

2.2 Separation Theorem

Theorem 3 (Separation Theorem). Let $C \subset \mathbb{E}$ be nonempty, closed, and convex, and let $x \notin C$, then there exists $s \in \mathbb{E} \setminus \{0\}$ with

$$\langle s, x \rangle > \sup_{v \in C} \langle s, v \rangle. \quad (14)$$

Substituting s with $-s$, we obtain

$$\langle s, x \rangle < \inf_{v \in C} \langle s, v \rangle. \quad (15)$$

To see the separation, define $\gamma = \frac{1}{2}(\langle s, x \rangle + \sup_{y \in C} \langle s, y \rangle)$. Then, $x \in \{z : \langle s, z \rangle > \gamma\}$ and $C \subseteq \{z : \langle s, z \rangle < \gamma\}$, i.e., x and C lie in two distinct half-spaces defined by the hyperplane

$$H = \{z : \langle s, z \rangle = \gamma\}. \quad (16)$$

3 Convex Functions

3.1 Definition and Examples

Definition 11 (Convex functions). A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is called convex if $\text{epi}(f)$ is a convex set.

We can substitute the the epigraph with strict epigraph.

Convex function also have convex level sets.

Examples of convex functions: scalar-valued linear functions $\mathcal{L}(\mathbb{E}_1, \mathbb{R})$ is convex; indicator of function of a convex set, any norm on $\mathbb{E}$, the scalar functions $x \mapsto x^2$, $x \mapsto \exp(x)$.

Proposition 2 (Domain of a convex function). *The domain of a convex function is convex.*

Using the projection linear mapping $L : (x, \alpha) \in \mathbb{E} \times \mathbb{R} \mapsto x \in \mathbb{E}$, we have $\text{dom}(f) = L(\text{epi}(f))$. We know that linear map preserve convexity.

Proposition 3 (Characterizing convexity). A function $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$ is convex iff for all $x, y \in \mathbb{E}$ and $\lambda \in (0, 1)$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (17)$$

We can extend the above characterization.

Definition 12 (p -dimensional simplex). The p -dimensional simplex is the set

$$\Delta_p = \{\lambda \in \mathbb{R}^p : \lambda_i \geq 0, \sum_{i=1}^p \lambda_i = 1\}. \quad (18)$$

Corollary 4 (Jensen's Inequality). Let $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$ be a convex function, and let $x_1, \dots, x_n \in \text{dom}(f)$ with $\lambda_1, \dots, \lambda_n \geq 0$ such that $\sum_{i=1}^n \lambda_i = 1$. Then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (19)$$

Definition 13 (Convexity on a set). A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is convex on a nonempty convex set $C \subseteq \text{dom}(f)$ if the inequality (17) holds for every $x, y \in C$.

We mainly interest in proper convex (sometimes lsc) functions $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$. We take the following notation:

- $\Gamma = \Gamma(\mathbb{E}) = \{f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\} : f \text{ is proper and convex}\}$
- $\Gamma_0 = \Gamma_0(\mathbb{E}) = \{f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\} : f \text{ is proper, convex, and lsc}\}$

Corollary 3. Let $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$, then TFAE:

1. f is convex
2. f is convex on its domain

We note that convexity on the domain is equivalent to convexity on the whole space because the inequality trivially holds when x or y is outside the domain.

We then introduce stronger notions of convexity.

Definition 14 (Strict/strong convexity). Let $f \in \Gamma$ and $C \subset \text{dom}(f)$ a convex set, then f is said to be

- strictly convex if for all $x, y \in \mathbb{E}$ with $x \neq y$ and all $\lambda \in (0, 1)$, we have

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y). \quad (20)$$

- strongly convex with parameter $m > 0$ if for all $x, y \in \mathbb{E}$ and all $\lambda \in (0, 1)$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) - \frac{m}{2}\lambda(1 - \lambda)\|x - y\|^2. \quad (21)$$

where m is called the modulus of strong convexity on C

Note that strong convexity implies strict convexity, but not the reverse.

Proposition 4 (Characterization of strong convexity). Let $f \in \Gamma$ and $C \subset \text{dom}(f)$. Then f is strongly convex with parameter $m > 0$ on C iff the function $x \mapsto f(x) - \frac{m}{2}\|x\|^2$ is convex on C .

3.2 Functional operations preserving convexity

Proposition 5 (Positive combinations of convex functions). For $p \in \mathbb{N}$, let $f_1, \dots, f_p$ be convex (and lsc) and $\alpha_i \geq 0$ for $i = 1, \dots, p$. Then

$$f = \sum_{i=1}^p \alpha_i f_i \quad (22)$$

is convex (and lsc).

If, in addition, $\bigcap_{i=1}^p \text{dom}(f_i) \neq \emptyset$, then f is also proper.

Proposition 6 (Pointwise supremum of convex functions). For an arbitrary index set I , let f_i be convex (lsc) for all $i \in I$. Then the function

$$f(x) = \sup_{i \in I} f_i(x) \quad \forall x \in \mathbb{E} \quad (23)$$

is convex (lsc).

Index does not have to be finite.

Definition 15 (Affine functions). A function $f : \mathbb{E}_1 \rightarrow \mathbb{E}_2$ is called affine if

$$f(\lambda x + (1 - \lambda)y) = \lambda f(x) + (1 - \lambda)f(y), \quad \forall x, y \in \mathbb{E}_1, \lambda \in [0, 1]. \quad (24)$$

Every affine function can have a unique representation $f = L + v$ where $L \in \mathcal{L}(\mathbb{E}_1, \mathbb{E}_2)$ is linear and $v \in \mathbb{E}_2$.

If the coefficients need not sum to 1 then it's linear. Linear function is affine and affine function is linear if $f(0) = 0$.

Proposition 7 (Pre-composition of with an affine mapping). *Let $H : \mathbb{E}_1 \rightarrow \mathbb{E}_2$ be affine and $g : \mathbb{E}_2 \rightarrow \mathbb{R} \cup \{\infty\}$ (lsc and) convex. Then the function $f = g \circ H$ is (lsc and) convex.*

This is not true in general, as the composition of two convex functions may not be convex.

4 Minimization and convexity

In an minimization problem:

$$\min_{x \in C} f(x) \iff \min_{x \in \mathbb{E}} f(x) + \delta_C(x). \quad (25)$$

If f is convex and C is convex, then we call this problem a convex minimization/optimization problem.

4.1 General existence results

Definition 16 (Coercivity and supercoercivity). Let $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$. Then f is called

- coercive if $\lim_{\|x\| \rightarrow \infty} f(x) = \infty$, which is sometimes called o-coercive.
- supercoercive if $\lim_{\|x\| \rightarrow \infty} \frac{f(x)}{\|x\|} = \infty$, which is sometimes called 1-coercive.

Lemma 2 (Level-boundedness and coercivity). A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is coercive iff f is level-bounded.

We now extent Theorem 1 to cover the problem above.

Corollary 5 (Existence of minimizer). Let $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{\infty\}$ be lsc and let $C \subseteq \mathbb{E}$ be closed such that $\text{dom}(f) \cap C \neq \emptyset$ and suppose one of the following holds:

1. f is coercive
2. C is bounded

Then f has a minimizer on C .

Proof: Consider the function $g = f + \delta_C$. Then it holds that $\text{lev}_{\leq \alpha} g = \text{lev}_{\leq \alpha} f \cap C$ for all $\alpha \in \mathbb{R}$.

Hence, under either assumption, g is closed and bounded level sets. Hence, it is lsc and level-bounded, Then the results follows from Theorem 1. $\square$

5 Differential Theory

5.1 Subdifferential

Definition 17 (Subdifferential). Let $f : \mathbb{E} \mapsto \overline{\mathbb{R}}$, and $\bar{x} \in \mathbb{E}$. Then $g \in \mathbb{E}$ is call a subgradient of f at $\bar{x}$ if

$$f(x) \geq f(\bar{x}) + \langle g, x - \bar{x} \rangle, \quad \forall x \in \mathbb{E}. \quad (26)$$

The set of all subgradients of f at $\bar{x}$:

$$\partial f(\bar{x}) = \{g \in \mathbb{E} : f(x) \geq f(\bar{x}) + \langle g, x - \bar{x} \rangle, \forall x \in \mathbb{E}\} \quad (27)$$

is called the subdifferential of f at $\bar{x}$.

Subdifferential contains exactly the slopes of all affine minorants of f at $\bar{x}$.

Proposition 8 (Elementary properties of the subdifferential). Let $f : \mathbb{E} \mapsto \overline{\mathbb{R}}$. Then the following hold:

1. $\partial f(\bar{x})$ is closed and convex for all $\bar{x} \in \text{dom}(f)$.
2. If f is proper, then $\partial f(\bar{x}) = \emptyset$ for all $\bar{x} \notin \text{dom}(f)$.
3. $0 \in \partial f(\bar{x})$ iff $\bar{x} \in \arg \min f$.
4. If f is convex, then $\partial f(\bar{x}) = \{v \in \mathbb{E} : (v, -1) \in N_{\text{epi}(f)}(\bar{x}, f(\bar{x}))\}$.

Theorem 4 (Subdifferential and conjugate function). Let $f : \mathbb{E} \mapsto \overline{\mathbb{R}}$. Then the following are equivalent:

1. $y \in \partial f(x)$.
2. $x \in \arg \min_z \{\langle z, y \rangle - f(z)\}$
3. $f(x) + f^*(y) = \langle x, y \rangle$ [Fenchel-Young equality]

If $f \in \Gamma_0$, these are also equivalent to

4. $x \in \partial f^*(y)$
5. $y \in \arg \min_z \{\langle x, z \rangle - f^*(z)\}$

Note that (1) and (4) are equivalent to ∂f^* is an inverse of ∂f .

Proof: Notice that

$$y \in \partial f(x) \iff f(z) \geq f(x) + \langle y, z - x \rangle, \forall z \in \mathbb{E} \quad (28)$$

$$\iff \langle z, y \rangle - f(z) \leq \langle x, y \rangle - f(x), \forall z \in \mathbb{E} \quad (29)$$

$$\iff \langle x, y \rangle - f(x) \geq \sup_z \{\langle z, y \rangle - f(z)\} = f^*(y) \quad (30)$$

Combine with Fenchel-Young inequality $f(x) + f^*(y) \geq \langle x, y \rangle$, we have

$$\iff f(x) + f^*(y) = \langle x, y \rangle \quad (31)$$

Apply the same argument to f^* and using the fact that $f^{**} = f$ for $f \in \Gamma_0$, we complete the proof. $\square$

5.2 Directional Derivative of a convex function

Definition 18 (Directional derivative). Let $f : \mathbb{E} \mapsto \overline{\mathbb{R}}$ be proper, and $\bar{x} \in \text{dom}(f)$. We say that f is directionally differentiable at $\bar{x}$ in the direction $d \in \mathbb{E}$ if the limit

$$f'(\bar{x}; d) = \lim_{t \downarrow 0} \frac{f(\bar{x} + td) - f(\bar{x})}{t}. \quad (32)$$

exists in an extended real-valued sense (so it can be $+\infty$). In this case, we call $f'(\bar{x}; d)$ the directional derivative of f at $\bar{x}$ in the direction d .

Remark: if f is differentiable, then $f'(\bar{x}; d) = \langle \nabla f(\bar{x}), d \rangle$.

Definition 19 (Positive Homogeneity). We call a function $f : \mathbb{E} \mapsto \mathbb{R} \cup \{+\infty\}$ positively homogeneous if $f(\lambda x) = \lambda f(x)$ for all $x \in \mathbb{E}$ and $\lambda \geq 0$.

Proposition 9. A function $f : \mathbb{E} \mapsto \mathbb{R} \cup \{+\infty\}$ is convex and positively homogeneous if and only if

$$h(\lambda x + \mu y) \leq \lambda h(x) + \mu h(y), \quad \forall x, y \in \mathbb{E}, \lambda, \mu \geq 0. \quad (33)$$

a property called sublinearity. Sublinearity is also equivalent to subadditivity $h(x + y) \leq h(x) + h(y)$ and positive homogeneity.

That said, convexity and positive homogeneity give subadditivity:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad \forall x, y \in \mathbb{E}, \lambda \in [0, 1]. \quad (34)$$

Set $\lambda = \frac{1}{2}$, we have

$$f\left(\frac{x + y}{2}\right) \leq \frac{f(x) + f(y)}{2} \implies f(x + y) \leq f(x) + f(y). \quad (35)$$

$\square$

Observe that any positive homogeneous function has $h(0) = 0$ and any sublinear function satisfies $-h(x) \leq h(0) + h(-x) \leq h(x)$.

Proposition 10 (Directional derivative of a convex function). *Let $f \in \Gamma_0$, $x \in \text{dom}(f)$, and $d \in \mathbb{E}$. Define:*

$$t \in \mathbb{R} \setminus \{0\} \mapsto q(t) = \frac{f(x + td) - f(x)}{t}. \quad (36)$$

then TFAE:

1. $q(-t) \leq q(-s) \leq q(s) \leq q(t)$ for all $0 < s < t$.
2. We have $f'(x; d) = \inf_{t>0} q(t)$. In particular, $f'(x; d)$ always exists in an extended real-valued sense.
3. $\text{dom}(f') = \mathbb{R}_+(\text{dom}(f) - x)$.
4. If $x \in \text{int}(\text{dom}(f))$, then $f'(x; \cdot)$ is finite-valued, positively homogeneous, and convex, (so sublinear).

Proof: (2): Since infima always exists in $\overline{\mathbb{R}}$, $\inf_{t>0} q(t)$ exists. By (1), $q(t)$ monotonically decreasing as $t \downarrow 0$, so we have

$$\inf_{t>0} q(t) = \lim_{t \downarrow 0} q(t) = f'(x; d). \quad (37)$$

(3): Follows from (2) we have:

$$d \in \text{dom}(f') \iff \exists t > 0 : \frac{f(x + td) - f(x)}{t} < \infty. \quad (38)$$

$$\iff \exists t > 0 : f(x + td) - f(x) < \infty. \quad (39)$$

$$\iff \exists t > 0 : x + td \in \text{dom}(f), \text{ since } f(x) < \infty \text{ for } x \in \text{dom}(f). \quad (40)$$

$$\iff \exists t > 0 : td \in \text{dom}(f) - x. \quad (41)$$

$$\iff d \in \mathbb{R}_+(\text{dom}(f) - x). \quad (42)$$

(4): Take $0 < s < t$. If $x \in \text{int}(\text{dom}(f))$, we can choose $t > 0$ sufficiently small such that both $x \pm td \in \text{int}(\text{dom}(f))$. Thus $q(t)$ and $q(-t)$ are finite. Hence $+\infty > q(t) \geq q(s) \downarrow f'(x; d) \geq q(-t) > -\infty$. As $d \in \mathbb{E}$ is arbitrary, $f'(x; \cdot)$ is finite-valued.

$f'(x, \cdot)$ is also positively homogeneous as:

$$f'(x; \alpha d) = \lim_{t \downarrow 0} \frac{f(x + t\alpha d) - f(x)}{t} \quad (43)$$

$$= \lim_{t \downarrow 0} \alpha \frac{f(x + t\alpha d) - f(x)}{t\alpha} \quad (44)$$

$$= \alpha f'(x; d). \quad (45)$$

Convexity of $f'(x; \cdot)$: for $t > 0, d, h \in \mathbb{E}$ and $\lambda \in (0, 1)$, by convexity

For notation:

$$\mathbb{R}_+(S) := \{\alpha s \mid s \in S, \alpha \geq 0\}.$$

$$d \in \mathbb{R}_+(S) \iff \exists \alpha \geq 0, s \in S : d = \alpha s.$$

of f we have

$$f(x + t(\lambda d + (1 - \lambda)h)) - f(x) = f(\lambda(x + td) + (1 - \lambda)(x + th)) - f(x) \quad (46)$$

$$\leq \lambda(f(x + td) - f(x)) + (1 - \lambda)(f(x + th) - f(x)). \quad (47)$$

which implies

$$f'(x; \lambda d + (1 - \lambda)h) = \lim_{t \downarrow 0} \frac{f(x + t(\lambda d + (1 - \lambda)h)) - f(x)}{t} \quad (48)$$

$$\leq \lim_{t \downarrow 0} \lambda \frac{f(x + td) - f(x)}{t} + (1 - \lambda) \frac{f(x + th) - f(x)}{t} \quad (49)$$

$$= \lambda f'(x; d) + (1 - \lambda) f'(x; h). \quad (50)$$

□

Definition 20 (Linearity space). For a sublinear function $h : \mathbb{E} \mapsto \mathbb{R} \cup \{+\infty\}$, we call the set

$$\text{lin}(h) = \{d \in \mathbb{E} : -h(d) = h(-d)\} \quad (51)$$

the linearity space of h .

Lemma 3 (Linearity space of a sublinear function). Let $h : \mathbb{E} \mapsto \mathbb{R} \cup \{+\infty\}$ be sublinear. Then the following holds:

1. The linearity space $\text{lin}(h)$ is a subspace on which h is linear.
2. For $\bar{x} \in \text{int}(\text{dom}(h))$ and $q = h'(\bar{x}, \cdot)$, we have:

$$(a) \quad q(\lambda \bar{x}) = \lambda h(\bar{x})$$

$$(b) \quad q \leq h$$

$$(c) \quad \text{lin} q \supset \text{lin} h + \text{span}\{\bar{x}\}$$

Theorem 5 (Subdifferential vs. directional derivative). Let $f \in \Gamma$, then the following holds:

1. For all $\bar{x} \in \mathbb{E}$, we have $\partial f(\bar{x}) = \{s \in \mathbb{E} : \langle s, d \rangle \leq f'(\bar{x}, d), d \in \mathbb{E}\}$.
2. For all $\bar{x} \in \text{int}(\text{dom}(f))$, we have

$$f'(\bar{x}; d) = \sup_{s \in \partial f(\bar{x})} \langle s, d \rangle, \quad \forall d \in \mathbb{E}. \quad (52)$$

In particular, $\partial f(\bar{x})$ is nonempty.

5.3 Continuity properties

The subdifferential operator of a closed, proper, and convex function has a closed graph.

Corollary 6 (Outer semicontinuity of the subdifferential). *Let $f \in \Gamma_0$ and suppose $\{x_k\} \rightarrow x$ and $\{y_k \in \partial f(x_k)\} \rightarrow y$. Then $y \in \partial f(x)$, i.e. the graph of ∂f is closed.*

Proof: By Fenchel-Young equality, we have $f(x_k) + f^*(y_k) = \langle x_k, y_k \rangle$ for all k . On the other hand, by lsc of f and f^* , we have

$$f(x) + f^*(y) \leq \liminf_{k \rightarrow \infty} f(x_k) + \liminf_{k \rightarrow \infty} f^*(y_k) \leq \liminf_{k \rightarrow \infty} (f(x_k) + f^*(y_k)) = \liminf_{k \rightarrow \infty} \langle x_k, y_k \rangle = \langle x, y \rangle. \quad (53)$$

$f(x) + f^*(y) \leq \langle x, y \rangle$. Together with Fenchel-Young inequality $f(x) + f^*(y) \geq \langle x, y \rangle$, we have $f(x) + f^*(y) = \langle x, y \rangle$. This implies $y \in \partial f(x)$. $\square$

A proper convex function is locally Lipschitz continuous on the interior of its domain.

Definition 21 (Locally Lipschitz continuous). *A function $f : \mathbb{E} \mapsto \mathbb{R} \cup \{+\infty\}$ is locally Lipschitz continuous at $\bar{x} \in \text{dom}(f)$ if $\exists L_{\bar{x}}, \delta_{\bar{x}} > 0$ such that*

$$|f(x) - f(y)| \leq L_{\bar{x}} \|x - y\|, \quad \forall x, y \in B(\bar{x}, \delta_{\bar{x}}). \quad (54)$$

Remark: every function that is locally Lipschitz at a point is also continuous at that point.

If a function is locally Lipschitz at every point of a compact set, then it is globally Lipschitz on that set.

Theorem 6 (Local Lipschitz continuity of a proper convex function). *Let $f \in \Gamma$ and $\bar{x} \in \text{int}(\text{dom}(f))$. Then f is locally Lipschitz continuous at $\bar{x}$.*

Definition 22 (Domain of continuity). *For a function $f : \mathbb{E} \mapsto \mathbb{R} \cup \{+\infty\}$, we call the set*

$$\text{cont}(f) = \{x \in \mathbb{E} : f \text{ is continuous at } x\} \quad (55)$$

the domain of continuity of f .

Corollary 7. *Let $f \in \Gamma$. Then $\text{int}(\text{dom}(f)) = \text{cont}(f)$.*

Theorem 7 (Boundedness properties of the subdifferential). For $f \in \Gamma$, the following holds:

1. Let $X \subset \text{int}(\text{dom}(f))$ be nonempty, open, and convex. Then f is Lipschitz continuous with modulus $L > 0$ on X if and only if $\|v\| \leq L$ for all $v \in \partial f(x)$ and $x \in X$.
2. ∂f maps bounded sets which are compactly contained in $\text{int}(\text{dom}(f))$ to bounded sets.

Corollary 8. Let $f \in \Gamma$ and $\bar{x} \in \text{int}(\text{dom}(f))$. Then $\partial f(\bar{x})$ is nonempty, compact, and convex.

Proof: By elementary properties of the subdifferential, $\partial f(\bar{x})$ is closed and convex since $\bar{x} \in \text{int}(\text{dom}(f))$. By the boundedness properties of the subdifferential, since $\{\bar{x}\}$ is compactly contained in $\text{int}(\text{dom}(f))$, $\partial f(\bar{x})$ is bounded. Hence $\partial f(\bar{x})$ is compact. $\square$

5.4 Differentiable convex function

Differentiability of a convex function is defined on an open set.

Theorem 8. Let $f \in \Gamma$ and $\bar{x} \in \text{int}(\text{dom}(f))$. Then $\partial f(\bar{x})$ is a singleton iff f is differentiable at $\bar{x}$. In this case, $\partial f(\bar{x}) = \{\nabla f(\bar{x})\}$.

Differentiability on an open set implies continuous differentiability on that set.

Theorem 9. Let $f \in \Gamma$ and $\bar{x} \in \text{int}(\text{dom}(f))$. Then f is continuously differentiable at $\bar{x}$ iff ∂f is a singleton for all $x \in \text{int}(\text{dom}(f))$ and the mapping $x \mapsto \partial f(x)$ is continuous at $\bar{x}$.

Corollary 9. Let $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex. Then f is differentiable iff f is continuously differentiable.

Second-order characterizations

Theorem 10. Let $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{+\infty\}$ be twice differentiable on the open convex set $\Omega \subset \text{int}(\text{dom}(f))$. Then the following hold:

1. f is convex on Ω iff $\nabla^2 f(x) \succeq 0$ for all $x \in \Omega$.
2. If $\nabla^2 f(x)$ is positive definite for all $x \in \Omega$, then f is strictly convex on Ω .
3. f is σ strongly convex on Ω iff for all $x \in \Omega$, the smallest eigenvalue of $\nabla^2 f(x)$ is bounded below by σ , i.e. $\nabla^2 f(x) \succeq \sigma I$.

Corollary 10. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{1}{2}x^T A x + b^T x + c, \quad (56)$$

with $A \in \mathbb{R}^{n \times n}$ symmetric, $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$. Then f is convex iff $A \succeq 0$, and f is σ -strongly convex iff f is strictly convex iff A is positive definite.

5.5 Subdifferential calculus

Theorem 11. For all $i \in I = 1, \dots, m$, let $f_i \in \Gamma$ and $x \in \bigcap_{i=1}^m \text{int}(\text{dom}(f_i))$. Set $f = \max_{i \in I} f_i$ and $I(x) = \{i \in I : f_i(x) = f(x)\}$. Then

$$\partial f(x) = \text{conv} \left(\bigcup_{i \in I(x)} \partial f_i(x) \right). \quad (57)$$

Corollary 11. For all $i \in I = 1, \dots, m$, let $f_i \in \Gamma$ be differentiable at $x \in \bigcap_{i=1}^m \text{int}(\text{dom}(f_i))$. Set $f = \max_{i \in I} f_i$ and $I(x) = \{i \in I : f_i(x) = f(x)\}$. Then

$$\partial f(x) = \text{conv} \left(\bigcup_{i \in I(x)} \{\nabla f_i(x)\} \right). \quad (58)$$

Theorem 12 (Fenchel-Rockafellar duality). Let $f : \mathbb{E}_1 \rightarrow \mathbb{R} \cup \{+\infty\}$ and $g : \mathbb{E}_2 \rightarrow \mathbb{R} \cup \{+\infty\}$ and $L \in \mathcal{L}(\mathbb{E}_1, \mathbb{E}_2)$. Then the followings hold:

1. (Weak duality) We have

$$p := \inf_{x \in \mathbb{E}_1} \{f(x) + g(L(x))\} \geq \sup_{y \in \mathbb{E}_2} \{-f^*(L^*(y)) - g^*(-y)\} =: d. \quad (59)$$

2. (Strong duality) If f, g are convex and the qualification condition $0 \in \text{int}(\text{dom}(g) - L(\text{dom}(f)))$ holds, then equality holds, i.e, $p = d$, and the supremum is attained in finite.

3. (Primal-dual recovery) If $f \in \Gamma_0(\mathbb{E}_1)$, $g \in \Gamma_0(\mathbb{E}_2)$, the following are equivalent for $\bar{x} \in \mathbb{E}_1$ and $\bar{y} \in \mathbb{E}_2$:

- (a) $p = d$, $\bar{x} \in \arg \min \{f(x) + g(L(x))\}$, and $\bar{y} \in \arg \max \{-f^*(L^*(y)) - g^*(-y)\}$.
- (b) $L^*(\bar{y}) \in \partial f(\bar{x})$, and $-\bar{y} \in \partial g(L(\bar{x}))$.
- (c) $\bar{x} \in \partial f^*(L^*(\bar{y}))$, and $L(\bar{x}) \in \partial g^*(-\bar{y})$.

We note that the qualification condition is satisfied under a stronger

condition $\text{int}(\text{dom}(g)) \cap L(\text{dom}(f)) \neq \emptyset$, which is often easier to verify in practice.

We call $\inf_{x \in \mathbb{E}_1} \{f(x) + g(L(x))\}$ the primal problem, and $\sup_{y \in \mathbb{E}_2} \{-f^*(L^*(y)) - g^*(-y)\}$ the dual problem. Part (2), (3) of the theorem say under the qualification condition, solving the primal problem is equivalent to solving the dual problem, and we can get the $\arg \min, \arg \max$ from (3).

Fenchel duality gives us the sum rule for subdifferential:

Theorem 13 (Generalized sum rule). *For $f : \mathbb{E} \rightarrow \mathbb{R} \cup \{+\infty\}$ and $g : \mathbb{E} \rightarrow \mathbb{R} \cup \{+\infty\}$, and $L \in \mathcal{L}(\mathbb{E}, \mathbb{E})$, the following hold:*

1. $\partial(f + g \circ L)(x) \supset \partial f(x) + L^*(\partial g(L(x)))$ for all $x \in \mathbb{E}$.
2. If $f \in \Gamma_0(\mathbb{E})$, $g \in \Gamma_0(\mathbb{E})$, and the qualification condition $0 \in \text{int}(\text{dom}(g) - L(\text{dom}(f)))$ holds, the equality holds, i.e. $\partial(f + g \circ L)(x) = \partial f(x) + L^*(\partial g(L(x)))$ for all $x \in \mathbb{E}$.

Corollary 12 (Subdifferential sum rule). *For $f, g \in \Gamma$, then*

$$\partial(f + g)(x) \supset \partial f(x) + \partial g(x), \quad \forall x \in \mathbb{E}. \quad (60)$$

If the qualification condition holds, equality holds, i.e. $\partial(f + g)(x) = \partial f(x) + \partial g(x)$ for all $x \in \mathbb{E}$.

Corollary 13 (Subdifferential chain rule). *Let $g \in \Gamma(\mathbb{E}_2)$ and $L \in \mathcal{L}(\mathbb{E}_1, \mathbb{E}_2)$. Then $\partial(g \circ L)(x) \supset L^*(\partial g(L(x)))$. Under the qualification condition, the equality holds, i.e. $\partial(g \circ L)(x) = L^*(\partial g(L(x)))$ for all $x \in \mathbb{E}_1$.*

Theorem 14 (Subdifferential of infimal convolution). *Let $f, g \in \Gamma_0$ and $x \in \text{dom}(f \# g)$. Then*

$$\partial(f \# g)(x) = \partial f(\bar{u}) \cap \partial g(x - \bar{u}), \quad (61)$$

for all $\bar{u} \in \arg \min_u \{f(u) + g(x - u)\}$.

Corollary 14 (Derivative of Moreau envelop). *Let $f \in \Gamma_0$ and $\lambda > 0$. Then $e_\lambda f$ is continuously differentiable with $\frac{1}{\lambda}$ Lipschitz gradient given by*

$$\nabla e_\lambda f(x) = \frac{1}{\lambda}(I - \text{prox}_{\lambda f})(x). \quad (62)$$

Theorem 15 (Moreau decomposition). *Let $f \in \Gamma_0$ and $\lambda > 0$. Then for all $x \in \mathbb{E}$, we have*

$$x = \text{prox}_{\lambda f}(x) + \lambda \text{prox}_{\frac{1}{\lambda} f^*}\left(\frac{x}{\lambda}\right). \quad (63)$$

Proposition 11. *For any proper, convex, closed function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and $g : \mathbb{E} \rightarrow \mathbb{R}$ is continuously differentiable, then*

$$\partial(f + g)(x) = \partial f(x) + \nabla g(x), \quad \forall x \in \mathbb{E}. \quad (64)$$

Corollary 15 (Primal-dual optimality condition). *Consider the problems*

$$P. \min_x h(Lx) + g(x)$$

$$D. \max_y -g^*(-L^*(y)) - h^*(y)$$

where $g : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and $h : Y \rightarrow \overline{\mathbb{R}}$ are proper, closed, and convex functions, and $L \in \mathcal{L}(\mathbb{E}, Y)$.

Suppose the optimal value of the primal and dual problem are equal, as implied by the qualification condition. Then, x is the minimizer of the primal problem and y is the maximizer of the dual problem iff

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \begin{bmatrix} 0 & L^* \\ -L & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \partial g(x) \times \partial h^*(y). \quad (65)$$

References